ActInf Textbook Group ~ Cohort 7 ~ Session 4 (Chapter 2, part 1) ~ 8/5/2024

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all right welcome back cohort 7 and all Andrew to you and just take it however sure all right um yeah so welcome everyone we're uh cohort 7 here discussing chapter two of the active inference textbook by friston par at all and uh here we're looking at the low road to active inference so uh as we saw in the previous chapter there was a kind of schematic showing uh there might be two general paths that that one can take to to reach an understanding of active inference so one of them is kind of starting from uh something like basium mechanics and statistics which is what the low road is all about that's what we'll be talking about today and then from the other angle one can approach active inference by starting with the free energy principle and General uh imperative for for organisms to uh act and perceive and so on in order to achieve homeostasis uh where possible and again we can reach active inference from there so I was just going to give a hopefully brief uh summary of of chapter two um starting from the first section uh it begins with the helm holian or perhaps Conan perspective of perception as unconscious inference as well as the basy and brain hypothesis and so with this framework we're treating the cognitive mechanisms of perception action planning and learning as Basi and inference problems as well as deriving a variational approximation uh to overcome the typical intractability problems which arise when attempting to compute exact basian inference uh those Concepts or something we'll be getting into later in the chapter so uh to start with there's perception is inference uh the authors claim perception is not just a bottomup process of turning sensory States or observations into internal representations of the outside as if it were a one-way process from from outside in uh despite the fact that this has been argued historically in various cognitive science Traditions they say instead it is an inferential process that combines top- down prior information about the most likely causes of Sensations with bottomup sensory stimuli so it's more of an inside out process uh we could liken this to comparing a prior hypothesis with the outcomes of an experiment which act as our confirmatory or disconfirmatory evidence for our hypothesis so um from there we can relate this idea of priors and observations or evidence uh using B's rule so uh bases rule we can we can break that down into these kind of component parts so we have our priors probability of X is the nomenclature that's used in this chapter more frequently used for describing continuous time active inference models uh and then we have uh our marginal probability of observations evidence U probability of Y and then these get related through a fun uh what called a likelihood model or function which is probability of Y given X or the probability of the evidence given your priors and then we can flip that through the equation and use incoming new evidence to update uh what would be called our posterior probability of X conditioned on y so our probability of pin States given the New Evidence um we're given a kind of a brief refresher of course this is this is basing in probabilities so we're dealing with probabil uh probabilistic reasoning so uh for anyone in the math group or or otherwise who was still getting a feel for some of the mathematics going on here it's good to familiarize yourself at least at first with some of the simpler uh ideas behind probabilistic reasoning including the the sum Rule and and product rules regarding like probability distributions sum to one uh there are no negative probability values maybe you know in the process of computing your final result you might have something like that but in any case in the final outcome no negative values and um as well as like marginalization those kinds of things so we're given um we're given a example of like for uh someone who's visually perceiving an object and they're trying to decide is it a frog or an Apple so it's kind of like our our latent uh State space is we're trying to figure out like we're just given two simple examples a frog or an apple and and that can be used to to illustrate a variety of ideas uh including you know they run through how to compute surprise uh based on you know if if one initially strongly believes they're seeing an apple but then they see this object jump uh that that gets uh you know incorporated into their beliefs and you know if our agent assumes that apples don't jump then we can see that there's going to be a lot of belief updating going on that might tell them like oh this is more than likely a frog um and we can see how that plays out ph's role we can see how um you know the the the observation or why is the jumping and then the x is the late state is it a Frogger and apple and we have our likelihood model with probabilities saying you know the agent's beliefs what what is the probability of seeing jumping given that it could be a frog uh probability 0.81 what's the probability of not seeing jumping given it's a frog is 0.19 notice those values SAR to one so it's that sum rule um and so yeah it's it's worth kind of wrapping your head around that particular example for anyone still learning the maths um we're seeing we see that um really what's going on with belief updating is calculating this quantity surprise which is measured in a kind of information theoretic quantity called NS um it's really just um the negative log of of um the marginal probability over observations uh so surprise effectively quantifies the difference between a prior and a posterior probability it tells us how unlikely an observation was and then from there organisms uh undertake something like optimization through basian inference where uh the model updates its priors to reduce this surprisal quantity so the probability of seeing a frog increases with that example example given before uh and then we can view this also as model comparison in the sense that you had a mile uh excuse me a model from before and now you have a new model uh that also incorporates the new evidence and those can be kind of compared we also get u a description of this the distinction between surprise and basian surprise which uh here they use uh cack leeler Divergence to um kind of they equate that with B surprise and so cobac Le leer Divergence is used in a variety of fields but it's basically it's it's a way of trying to kind of quickly uh score the difference between two different distributions so you can see that you know here's what your beliefs were before here's what they are now coack le blur Divergence to like try and figure out how much belief updating uh had to happen to reach uh the new model so um we're also given more details on active inference and the significance of biological inference and optimality biological plausibility of computations uh the the concept of predictive coding which is is not necessarily common place but it's it's a very uh uh welld discussed uh uh model of something like perception and other kinds of processes going on in the brain for Neuroscience um and algorithms like predictive coding implement this more General Basi and inference scheme and they're Central in a lot of modern biological treatments of of perception and this kind of top down and bottom up kind of bidirectional scheme that we're discussing here um with optimality basian inference involves optimizing the cost function for those who are uh familiar with kind of machine learning nomenclature here the cost function is variation free energy considers the full distribution over in States rather than just point estimates like the mean or using uh uh maximum likelihood estimation uh and so the idea is that with to make not not just is is variational inference a way of overcoming intractability but but it's also relatable to the notion that biological beings have limitations they have you know a limited amount of of resources at their disposal like cognitive metabolic and so on so they rely on approximations which here we might equate with you know the idea of computing a variational posterior when using bases rule uh and this can be based for example on a mean field approximation which we'll see later in chapter 4 or there are other methods for for computing that um from there they move into action as inference so it's not just perception as they noted at the beginning of the chapter perception and planning and learning and and action they're they're all kind of working together as distinct cognitive functions but they're they're effectively doing the same thing they're all working towards minimization of free energy so agents not only change their minds through perception they also act on their environment to change it to influence the observations that the environment generates so that the agent can encounter less surprising observations uh that's in line with something like realizing their preferences or priors over observations that they hold so we're still again we're still looking at basis rule we're still talking about observations as like evidence that probability of Y uh value um and then this gets taken up into later discussions over the differences between variational free energy and expected free energy uh whereas we had variational free energy which takes into account past observations and current observations expected free energy will also take into account something like future not yet observed observations right and so there's this kind of this that plays into the notion of planning or planning is inference uh where where similar computations are carried out but it's it's projecting into the future so we introduced the idea of policies which are like sequences of actions an agent can take it's kind of like um you know an agent can come up with these kind of counterfactual scenarios if I do this policy versus this policy versus this policy you know what will be the outcome what will be the outcome in relation to what I want or rather what I expect which is that that uh prior over observations uh and then also we can break down that expected free energy term into a variety of of smaller terms that have this kind of value of being interpretable so expected free energy can be broken down into Information Gain and pragmatic value which relates to this kind of classic explore exploit dilemma um where where where in active inference uh exploration exploitation are actually incorpor into the same cost function you don't have to Define them entirely separately or apply any kind of arbitrary rules to determine what defines an action or an observation or otherwise as as in like Information Gain related or pragmatic value related so they're both there um and finally at the end of the chapter there's uh just kind of some general ideas that are uh thrown out there uh for consideration you know again a recap to uh variational free energy which is an upper bound approximation to surprise and can be minimized efficiently using chemical or neuronal message passing and information that is available to the organism's generative model uh we'll see a lot more on neuronal message passing in chapter five uh there's also this kind of more philosophical idea that maybe an agent is isomorphic with its prior including it's priors over observations it's like it's trying to to realize these beliefs that it has uh finally uh in the next chapter after this there will be kind of a flip on the the road to active inference it'll be the high road and it'll instead start from the free energy Principle as opposed to B's role I think that summary was long enough so I'll wrap it up there um but now I'm just going to open the floor to any questions or anything anyone found interesting about this chapter or otherwise yeah I'll jump in on a couple of things I took some notes just reading through um just wanted to clarify a couple of things as well so I looked into um the consideration of like forward mode so consider a feed forward is it considered like a feed forward kind of like model in terms of our projection of our expectation or an agent's expectation in terms of you know implicit bias versus an updating of that model through interaction and then I kind of linked that together with an agreement versus an argument which could end up with like a zero sum outcome as opposed to the agreement which gives you a range of agreement within that argument space um so I kind of considered it as a correlation from external sensory and potential unknown expression of inference as a coupled field of connected conscious which is getting a little bit into the esoteric but in terms of like the external summed with internal States or is it more of a coupled state that has the properties of both continuous and discret in certain cases so where you're linking the or the cross correlation with the X and the Y variables looking back to figure 2 two for example yeah and so so so big questions there I mean the and I'm sorry if I'm sorry if this is backtracking but um just some things that come to mind mind in relation to what you said at least is um you know as far as it being a a feed forward model like there'll be a lot more detail whenever message passing is brought up later and so there's there are these kind of forward and backward connections depending on which algorithm you kind of choose for minimizing free energy so you know it's it's almost like the the agent because they engage not just in you know pulling from memory and present observations but also uh expected observations in the future there's a kind of like it's as if the the future influences the present and then and then also the the past influences the present and that that's kind of just thinking about it as as message passing on a on a graph that has a temporal Dimension to it it's almost like you can imagine these two nodes that are passing messages like the past to the present and the future the presence so there's also a kind of like retrospective um aspect to to the way that this plays out and then um with figure 2.2 you know that that what that is doing is just trying to describe um how you know the there's this common phrase the map is not the territory so there's no assumption here at all that the the model equates to the environment and how the environment functions around it like you know that an example could be like a I have this in my notes it's a little silly but it's like a you someone watching a magician and the the true explanation for the the illusions that are being produced are far different from just saying like oh it's magic you know but the audience very well might believe oh it is Magic right so um that that's kind of their get hypothesis or explanation for what they're seeing their observations that are coming in despite the fact that it's it's probably not at all you know equivalent to what's really going on right um so that you know that that can play into more realistic scenarios where someone very well might be trying to do something a task and they might solve a problem and the truth is they might not really understand it and yet what they have learned to do and the model the generative model they do carry nonetheless could end up being successful at solving the the problem even though they're not really getting so um and that can play into you know active inference has been used for like uh studying Psychopathology for example so people experience delusions and you know things like that that that don't equate to the true environment around them but nonetheless it's you know playing out in their belief dating in their beliefs um you know as they speak to that one point um in terms of like the agreement when you have another agent that agrees right if you are both biased in that negative manner without any external you know way to unbias yourself right even if you haven't encountered or your model hasn't encountered another agent won't it then Reas and reinforce so you'll have a regression so like how how can we account for a model that regresses into that negative log and kind of stays in like a [Music] loop yeah I mean it's you know there there might be some I think that there's already existing active inference literature that tries to look at like you know agents who are communicating with each other um there's a bird song example that comes up just in this textbook alone but I mean there's also a really nice paper on that point um it's a little bit more social sciences oriented but uh couple folks on the scientific Advisory board with the Institute um were involved in publishing and yeah it's called epistemic communities under active inference and it's kind of you know uh they set up a simulation with a you know actual active inference agents who um they they communicate with one another another and share the their beliefs and it's it's purely just their beliefs like there's no reference to that direct outside world and what often happens uh depending on how you set up the agents and what their potential belief States can be and the observations that can can come into contact with it's like it's really interesting because you can see like their their opinions about what's going on they there's a kind of a you know a bias towards uh um kind of reconfirming their own beliefs and so if they hear opinions from another agent that match their own there there tends to be that kind of loop of like okay we both agree with each other and we just continue to more and more strongly uh reconfirm each other's beliefs right it's almost like it kind of gets worse if they have a distorted belief of what's going on and it's a really good simulation because it's it's like a multi-agent one right so so you can have agents who have their own like what what can happen is you can have many agents certain ones agree with each other other ones don't and suddenly you have these like polarization Dynamics you know where they they start to this group over here strongly believes a certain idea or have a certain belief and reconfirm it with each other whereas this group over here is doing the same thing but about entirely different yeah Bel so that there can be that kind of kind of looping aspect to it it's very relatable to the kind of Dynamics we see in like social networks and how people use Twitter and tweet and you know there tend to be these kind of small um you know in groups that form and that's the idea of an epistemic Community because you could use it to describe like you know a whole scientific field of research where you know there are these certain scientists in this particular subme and they just kind of all talking to each other and may or may not actually be in B other fields okay thanks yeah that's kind of that kind of covered what I was trying to look at there I guess also like I wanted to tie it back to would that still be considered a hidden State then because you know depending on the size of the group because I mean if if the inference of the majority is incorrect and the smaller group is correct right based on factual evidence despite the larger group not agreeing to it I mean the the ramifications are that the larger group is still most likely going to be the outcome in terms of how that model is move forward with right yeah I mean yeah just just point blank like yeah you know this is like at least like from a discreet model perspective um you know often times The partially observable markup decision making processes are used to describe agents and model individual agents so you know again with this this figure that Daniel has here it's just you know despite the as external observers like ourselves who were like setting up the simulation so we know this kind of true state in within the simulation it's like still the agents don't know it right it's a it's a partially observable setting so at the end of the day everything is just latent States or beliefs about latent States um so so in terms of like that as transparency versus opaqueness would it be periodically transparent in that case or would it remain just like semi-transparent overall it would remain yeah yeah it would remain semi-transparent I mean you you could you could set it up differently you could just do a regular markup decision- making process where it's like rather than the agent receiving definitively observations you could just say that they receive the true state of the environment in which case it'd be fully transparent right um yeah those are the kind of two paradigms that are used for discrete um discrete time models um and you could have similar things with doing setting up continuous time models and I I did want harking back sorry you did ask something regarding continuous versus discreet or if there's ever like a combination of the two and uh later in the textbook especially I would have a look at Chapters maybe skim chapter six but especially look at like chapter seven and 8 that's where you get more details each of those are dedicated to discret time models and continuous time models but then there's also these ideas of like hierarchical models where you know perhaps on the the lower level of the hierarchy there's a continuous model say it's uh you're modeling an organism that's taking in you know real time observations and maybe they're uh you there are different like physiological signals going on or be related to neuroimaging data or you know heart rate things like that but then on a higher level it could be something like more of a sort of cognitive belief related level like maybe you're trying to model like an agent who's moving around in real time physiological you know uh Dynamics but then also they have you know beliefs about what's going on that are actual tens it kind of categorical representations of what's going on at the at that higher level it could be a discrete model it's like oh based on you know X Y and Z on this real time lower level what do I think is going on you know at this this kind of higher order uhr level awesome thing to interject because I was curious about that H State yeah um so for that case you're talking about the observations or the sensors that the agent is itself like taking on um that it's observing different distributions of like observation input that it's taking in and so its internal model is actually like multimodal I guess you could say or um there's like four or five different models that you're just like going across and it's like uh I observe color and like my sensors are continuous all that is like a continuous signal and then I also observe ones and zeros from I don't know an ADC maybe I'm not human um and so like that has a very different distribution between is is that what you were talking about in terms of um the observations yeah I would say somewhat uh or I want to say yes but I want to make sure that I understand kind of the nuances um so first of all yes like you can definitely have uh like an agent who's basically taking in like multimodal sensory information in fact the the nomenclature in the textbook is there are different op observation modalities is how they kind of that's how they term that and so um the three kind of common observation modalities they account for are interoception exteroception and proprioception um these are really commonly used in like neuroscience and cognitive science literature like completely you know you don't necessarily have to do anything with active inference to to find these terms in the literature but uh so in setion is kind of like observations coming from within um you know it kind of goes beyond the bounds of what we might think of as like oh we just have five senses right but to to categorize it it's like interoception would be signals yeah coming from within you know someone notices that they're hungry or something um they could they could notice that exteroception would be senses like sensory information coming from without um you know so things that you're seeing from outside of yourself or or or things that you're feeling externally and finally proprioception is that relates to like physical space and movement so you can think of like like a like a reflex arc like a movement of your your arms uh for example and uh I think the the main distinctions between those three modalities it's I think it's just useful depending on your use case like your object of study you're looking at and then you know there are kind of patterns in those kind of data and ways of modeling them to where suddenly do become like pragmatic like instrumentally relevant to to to distinguish them like that um and then yeah it's you yes uh you can have an agent who like you know they come into contact with things that are you know binaries like ones and zeros as opposed to like a a continuous value like a heart rate or like a particular shape you know it's like it's like you can take the color red and put it on this really broad continuous Spectrum but then you can also say oh it's red not blue right so those kind of D yeah playing to putting together an active inference agent yeah one related point there is like these different distributions so here's 6.2 here's the underlying latent cause and then that can lead to all these different kinds of chains of causation so depending on what the situation is or like the need for modeling and then those different distributions could have different priors on them so for something that is maybe more survival loaded like uh uh certain prior prior preference playing the role of of the pragmatic value for like blood pH so policies are sought to normalize that at at the expense of like learning opportunities whereas if you're looking at like iade the icade might be driven on a given time scale by purely visual ambiguity so different distributions like just different ways and that's kind of the flexibility of the overall framework to be able to to show up at these materially different kinds of processes and have a unified approach and compose across them so I guess to speak to that too like I had a question related to this in terms of like how much is our sensory perception actually not free energy in terms of I mean if you have a disability you know with uh might be raw cone deficiency right where you're not having the same kind of um color perception but the same thing with our own Dynamics in terms of what we sensory we are sensory being but our Consciousness may not necessarily be right so like what we can detect within a given range of just taking the light spectrum right is to the high infrared or the Low Infrared and Beyond to microwave or ultraviolet but beyond that spectrum is the majority of the latent space where we might have sensory disposition but we just don't understand it yet so I'm just wondering how much of our sensory disposition as a whole actually reduces or increases rather um our expenditure of energy in the modeling and how it might skew the data you know given new parameters for sensory cognition if we find you know new data in the future yeah that's interesting like why can't you just believe that you're seeing outside of the visual Spectra it's like it might just be a jump too far because there's the absence it's not it's not just the absence of confirmatory or disconfirmatory but it just it's it's it's like there's no packets coming in that channel but so there' be no way it would be as as um fabricated as like a total imagination overlay on the visual Fields but be it'd be a freewheeling but but then what if it were like you were looking through an infrared filter and you really got to know an area and then like you took off the filter for a second and there was like a quote after image right the length of that after image would be about kind of the the um durability of the generative model on those nonvisual Realms I'm also thinking in terms of like new tool creation right where something would be testable so I mean if we get Quantum to a state where you know just data in itself is exchanged you know at a rate that's theoretically faster than the speed of light right that would have to change our perspectives on a lot of the way that we model things right in terms of time especially especially if we're dealing with discreet or the opposite that's one of the reasons why I guess I'm kind of just going a bit too far beyond I don't want to jump too far out but in terms of like time domains especially if you're jumping through different models that are classifying and reordering things differently when you need a taxonomy that's similar or flexible in terms of being able to take on you know a context free kind of formality that is still able to express the detail without utilizing more energy yeah I was try s sorry yeah I was trying to think of like an example that was kind of paradoxical in my mind and it might just be my own approach to it and it's something that I want to try which was like I'm I'm calling it like a method of zebras where if you're classifying a zebra as a horse is the zebra black or is the zebra white you can do you know variational distribution of if it's white or if it's black does it still necessarily make the zebra a horse and to what degree are all zebras white or all zebras black and then how would you reclassify it under that you know categorization scheme yeah so so there's a you know there's a concept of a structure learning it's surely not just used in active inference but but it is used in active inference and the idea is is like it not just for a model to update like these kind of not just update its beliefs in terms of the final you know values that are found within its the probability dis distributions describing its beliefs but also to actually modify the general structure of its model in sense of like oh introduce a new category that you didn't you know previously know existed or something like that you can kind of relate that to the idea of like uh uh something like sort of rewiring neuronal connections and forming new connections that allow for just new not just new information but like new kinds of information almost to be formed um and in that case you could do something like basy and model comparison for example and use like you know the AK Criterion or oasian information Criterion yeah it would just be a matter of comparing models from there and then uh Within in both variational and expected free energy uh we whenever we look at how those uh are broken down into like smaller subterms that are being like subtracted or added from each other Etc um that's kind of that interpretable aspect of of those terms it's nice it's like uh variational free energy can be also be recomposed as like complex uh negative complexity minus accuracy so they're both or excuse me plus sorry trying to pull that one the point is to to minimize complexity of the model while maximizing accuracy which I think kind of gets at one of the points that you made right so that the idea would be to part of free energy minimization is to try and keep your your model as simple as possible while keeping it accurate so that kind of tradeoff does play out both in structure learning and in the more General belief updating that we were discussing earlier maybe one make about this kind of getting getting to active inference along the low road it's going to come together in chapter four when we see the tools of chapter 2 deployed under the kind of umbrella of chapter 3 free energy principle so one one reading of this chapter is like for different cognitive phenomena different cognitive apparatus like different matches or mismatches that or or degrees of freedom with different agents and and Niche it's like whether it's was more like an inference or more like an action from a given side of the blanket this partition is going to enable a lot of flexible modeling so it's like just saying a large variety of processes are going to be amenable to being partitioned this way because some of the special cases include like full observability and then there's the whole Continuum of all the way on through total non-observability like I'm flipping a coin in a room that's the only date I'm getting and then there's another random number generator outside so it might be like a hopeless optimization from like actually explaining the V but that's kind of that's the special case that then you can rise and describe out of and the extent that it's possible to infer like pation is the sparsity of that graph so by kind of starting with a with a general case then a variety of of total to Total absence of different kinds of arbitrary distributions can be framed and any any which of those situations they're framed in a way that's like going to fit into this approximate basian computation setting so in terms of that too like I had another question that's kind of related to that um and it dealt with I can't I didn't write down what section it was under but um it dealt with um give and take in terms of um external factors um so how much can we consider to be optimization versus mimicry of an external factor that is given versus taken if the mimicry is misunderstood or suboptimal copy of a realistic measure so if the mimicry is based on external external or internal bias um that is agreed to to be true despite the unknown falsehood in the data would would that still be considered positive that would you know flip that negative logarithm back over or would it kind of stay in that kind of that Loop that we kind of talked about earlier in that sense I we might be almost ranging into the the the philosophical here um which there are plenty of conversations to be had from a more philosophical angle with that temperence but uh as far as I'm aware it it's still we're looking at like a AR like like basically the the true state is is still hidden it's just like the beliefs that the agent has like in their head so to speak or you know represented in some kind of way um yeah there's there's no so so from that angle active inference it's like we're always kind of dealing with you know potentially it's almost like the observations themselves are like metrics that we're using to try and measure the the probability of hate state right so it's that's um well I guess give one other one other connection just to the kind of again this like agreement or like negotiation argument continua so this is like think about it from the Blanket's perspective so here's the niche here's Davis and the summer here's you know what I prefer to see pragmatic value for for my Thermo receptors so the thermo receptor could be cleaved off if this was all you saw you might think wow well the thermo is getting cleaved off on the right side but it's getting cleaved off on the left side too so that's just sort of a a convention because the interface is by definition part that's shared SL overlap so then there'll be a situation it's like wow we both are you know tracing the same path then that' be a situation where like the pragmatic values align that could be like an improv setting where it's just like okay we're going to do this now so it's like we're setting another set of rules on either side of the blanket that are making the movement of Y more coordinated but then if it was like I want it cooler and the environment wants it hotter then that's like the change the world change the mind but those those are going to have a a fundamental like one getting more accurate will make the other one less accurate but then you could have other situations where the there's like where why could be more accurate together than in inert one on the other side yeah that makes sense I guess to bring it like because yeah it is kind of a little bit philosophical so like to bring it back to more of a scientific standpoint like in terms of and I know this is kind of Leading Edge stuff as well but in terms of like increasing neuroplasticity um you're are you going to retain that information so if you can you know bring your own plasticity up in terms of you know neuronal firing or activity in the brain is that going to remain as an expanded state in terms of like a a library state for example that you can draw upon um as opposed to you know what we were just talking about in terms of like the loop of you might be thinking that you know that space is being increased but really what it's doing is actually reducing that space overall because you're also not allowing the priors to update that space so in terms of plasticity and especially with people that have lost plasticity so like Alzheimer's for example if you can bring that neuroplasticity back Can you reactivate those regions to bring old memory back in or would it have to be retrained like a model would be retrained yeah good questions substrate of memory open questions yeah there I'm I'm sure there's someone somewhere that I'm I'm not familiar with who's who's expressly doing research on Alzheimer's potentially from a an active inference perspective but yeah I mean all I I can say from the active inference point of view is that I mean neuroplasticity presumably is going to be guided by free energy minimization right so that there will be that that kind of consideration because no one's actually considering anything but uh you know the kind of consideration of like the complexity versus accuracy and the formation of new neuronal connections as well as like kind of modulating those by way of neurotransmitters like the the the kind of assumption made in active inference is that yeah it's just going to be free energy minimization and then of course with with that kind of that kind of deterioration that is actually leading to more difficulty for an individual to say maintain homeostasis or the kind of Life they want to live presumably there's some kind of external element that you know unfortunately has made it way in or otherwise that that that's you know it's as if the the neuronal connections and so on are trying to they're trying to keep going despite that you know occurring so they'll play into the process of the deterioration you know the same way that one tries to fight off a disease or something um but yeah don't know if that's clarifying or not but that's that' be my yeah no it is like it it I was just thinking back to I don't know if it was a stream from this week or last week Dan you might be more familiar but I dealt with the brain wave principle I think Carl was there with a couple other people um and I was thinking Robert Warden yeah I was thinking in terms of you know the brain wave um that would be message passing as in terms of like free energy where you're receiving where you might not be conscious of it right but if you have that activity or that space where you know it's a latent kind of memory that might crop up you know with a different experience it's like how that interact ction might be considered to skew the data or would it actually bind the data into a more calculable space in terms of prediction and um message passing and data data sharing cross modulation Etc there's essentially never a directional answer in the general case it just is a rare situation I'm I I think it'd be fun to think of some but it's almost always possible to think of niches for which any given like anything could be the other way right so the whole point is it it doesn't have a meta prior on this or that which gives you the expressibility but nothing's going to come along for free into the structure generative model David blumen today gave a really interesting Twist on this with training neural networks to approximate actm so that's this angle and his his argument was like at some level you're going to have to do something like machine learning based parameter sweeping optimization like it if you're just building it pedagogically you just want to see the the gears turn yeah you might do a kind of a custom curated Soft Choice of parameters or you might engineer a second layer optimization system but at some point you're going to have to have that Kel of the generative model and the more strongly typed thing it's going to be surrounded in a soft layer of cognition whether you go custom craft or whether you do some other kind of statistical method and he's like it's an approximation anyway you're approximating analytically it's already an approximation so if you're if you're approximating an approximator and in practice the way you approximate it really matters why not lean into it and just go for it and treat it like an engineering problem where it's just purely empirical almost anything you can think about I mean doesn't that lead into a quagmire of like back into like its Turtles all the way down though like the more refined you go like it just descends into more necessity forment the that that is if you think it's structured all the way it's like well it be a CH a markof chain of turtles but if it's like what if it was like well I see two turtles trailing off into the fog and then it's like one person says I think they Trail on forever and someone else says I just don't know that's just where that's where the the information supply chain I just don't know and I I can't tell any sensory evidence to differentiate these hypothesis Beyond this Horizon so at some point the statistical model is going to blur into the real world it's just cartography and so at that point you have to accept that it does blur into the real world yeah I just was wondering like how much in terms of like how much of that blur becomes you know actualized or does it become like almost forced and imposed upon reality right through the predictive nature of like a sterile environment like at a certain point there has to be like a reality check that might just kind of not break the model but might Hind the model in a way where it doesn't become as effective right because any of us may not have perceived you know that that random outcome for example you could make a niche where a model's performance was strongly tied or you can make a model in Niche where the outputs were not so you'd have to give a lot more information about what you're talking about but that's why it's like you're you know opening up a big Vista then specking it out within that right I mean the narrow path is the more difficult to begin with right but that's usually the more fruitful it's like rather than the low hanging fruit you know how you getting to the top to get to the best kind of fit but my question I guess there is like at what point does that become more risk than is required of you know the model and does that then interfere with free energy as a whole it's I guess it goes back to just the basis of the tradeoffs but yeah it's again that like plastic you know complexity versus accuracy problem can just sorry I'm also looking at a question that was asked in the chat so like at a point a model becomes so complex that it hinders its performance and it's like yeah that that can that can happen uh it might be that the agent is dealing with a really complex you know problem then over time it just it figures out you know here's one very precise solution to the problem in a particular instance but then it's just really heavily overfits to that particular problem like that's that's the one solution it kind could come up with and then it's not particularly robust so as soon as the environment changes just a little bit suddenly that solution that hypothesis or set of beliefs that the agent has just like completely you know falls apart in terms of performance or you know accuracy or something along lines that on framework as well though go ahead wouldn't that be dictated on like the solidity of like the framework itself that you're building off of like how much of the structure will collapse into itself versus what will remain that's still salvageable in terms of re-up dating yeah it' be kind of like having bad priers right like that this is this is from once you are like you know planning and and you know so so it's like an agent who has their beliefs in the present and their prior you know information that they're working with and then once they start planning into the future if those priers are you know completely terrible so to speak or inaccurate at least relative to what's going on around them then it's like yeah you're going to come up with some some pretty pretty faulty plans as well right so broader projected hypotheses you're making are like you know also going to be bumped yeah it also leads to more reactivity I don't know if anybody's seen the I think it was a study on like I think it was like the hungry judge study something like that where they did a study of Judges that made rulings before lunch and after lunch and the accuracy I thought that was interesting I don't know if any anybody has run across that I'll have to check that classic stolski Contra will yeah um okay thank you Andrew um so next week we'll stay in chapter two or come at this time and we'll be in chapter 7 chapter 7 I mean it's it's revisiting the low road it's going through a specific kind of architecture so two and seven read fine together again interestingly three and eight with the continuous time in certain ways having more Affinity with the free energy principle path formulation but uh either way another week in these chapters then third week possibly Andrew will do the social science model dream and then maybe in the applying we we could we could do some tweaking um and and and adding different things like to the script and notebook version of your social science simulation and then in the third week here we can um we we could do we can we can see see where we're at then and uh yeah thank you all thanks thank you Courtney thank you Zay and Dave and everyone see you nice to meet you peace bye later thanks thanks